AgentsKG: A Hierarchical Multi-Agent Framework for Open-Domain Knowledge Graph Construction

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Abstract

Knowledge Graph Construction (KGC) is essential for transforming unstructured text into structured knowledge representations. Despite advances in Large Language Models, existing methods treat KGC as a single-pass generation task, conflating extraction, normalization, and validation within a single forward pass. This leads to hallucinated facts, polysemous conflation, and fragmented triples, particularly in open-domain settings where predefined schemas are unavailable. In this work, we propose AgentsKG, a hierarchical multi-agent framework that decouples semantic perception from structural integration. In the Semantic Perception Layer, a multi-role Verification Committee filters hallucinated and invalid assertions through majority voting, while a Contextual Profiler resolves polysemous ambiguities by enriching mentions with context-dependent semantic descriptors. In the Structural Integration Layer, a Knowledge Linker merges redundant entities and relations based on semantic profiles, and an Ontological Logic Auditor enforces logical consistency across the graph. Extensive experiments demonstrate that AgentsKG outperforms state-of-the-art training-free baselines in both extraction accuracy and structural quality, offering a robust approach to open-domain knowledge graph construction without additional training. Source code is available at https://doi.org/10.5281/zenodo.20484211

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KDD '26, Jeju Island, Republic of Korea

ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3818058

CCS Concepts

• **Computing methodologies → Information extraction**; *Multi-agent systems*; **Semantic networks**.

Keywords

Knowledge Graph Construction, Multi-agent Systems, Large Language Models, Knowledge Graph Canonicalization

ACM Reference Format:
Shilong Liu, Yongqiang Liu, Jiye Liu, Xuan Guo, Tianpeng Li, Qiyao Peng, and Wenjun Wang. 2026. AgentsKG: A Hierarchical Multi-Agent Framework for Open-Domain Knowledge Graph Construction. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3770855.3818058

1 Introduction

Knowledge Graphs (KGs) provide a structured and interpretable representation of real world knowledge, serving as a foundational component for knowledge-driven artificial intelligence systems [14, 17]. By organizing facts into symbolic triples of the form $\langle s, p, o \rangle$ [27], KGs allow machines to store, query, and reason over information in a transparent manner. Such structured representations are critical for a wide range of downstream applications, including knowledge-augmented generation [20, 39], explainable reasoning [22], and decision support systems [46]. However, the practical utility of these systems is fundamentally governed by the precision of Knowledge Graph Construction (KGC) [35]. Hence, KGC research has gained increasing popularity in industry and academia. As information grows exponentially, KGC has become the pivotal bridge that transforms noisy raw text into a queryable and consistent knowledge backbone.

To meet the growing demand for large-scale KGs, KGC has increasingly moved from manual curation to automated pipelines [45]. Early approaches relied on task-specific models with hand-crafted patterns or supervised learning [2], but are parameter-bound and require retraining for new domains. LLMs have since introduced a prompt-driven paradigm [34, 37], allowing triples to be extracted from text without annotation or fine-tuning. Despite their flexibility, both lines of work treat KGC as a single-pass generation task[18]. That is, extraction, normalization, and factual validation are all handled within a single forward pass, yet these objectives are inherently in tension: high-recall extraction favors over-generation, while normalization and validation require conservative filtering. Conflating them in one pass often leads to hallucinated facts, conflated polysemous mentions, and fragmented triples that lack global coherence. Accordingly, there is a growing interest in multi-agent paradigms, where these intertwined objectives can be decomposed into collaborative sub-tasks handled by specialized roles.

Nevertheless, designing a multi-agent KGC framework for open-domain settings introduces its own challenges, which can be characterized along two dimensions.

Challenge 1: Semantic Ambiguity. Without predefined type constraints, the extraction stage is susceptible to two categories of semantic error. The first is polysemous conflation: identical surface forms may refer to distinct concepts depending on the context. For example, "Apple" may denote a technology company in one sentence and a fruit in another, yet a standard extractor would produce the same entity string for both [7, 32]. The second is factual hallucination: an LLM may assign an incorrect nationality to a public figure due to training bias, introducing noise that is not grounded in the source text. Without context-aware agents, both types of error can silently propagate in the graph and degrade its reliability [11].

Challenge 2: Structural Inconsistency. In the absence of a fixed schema, maintaining global coherence across the graph becomes non-trivial, and the difficulty escalates at multiple levels [13]. At the entity level, the framework needs to recognize that "Barack Obama" and "the 44th U.S. President" refer to the same node. At the relation level, synonymous predicates such as "was born in" and "birthPlace" should be aligned to avoid redundancy. Besides, logical contradictions may also arise in triples. For example, if both ⟨A, father_of, B⟩ and ⟨B, father_of, A⟩ are extracted, the inconsistent one should be identified and removed [8]. These challenges place concrete requirements on the design of multi-agent KGC frameworks: agents need to jointly address instance-level semantic grounding and graph-level structural governance.

To address these challenges, we propose AgentsKG, a hierarchical training-free KGC framework that assigns specialized agents to handle semantic perception and structural integration respectively.

For semantic ambiguity (**Challenge 1**), we design a group of agents that collaborate within a Semantic Perception Layer. A Verification Committee, composed of a KG Auditor, a Domain Expert, and a Fact-checker, cross-examines each candidate triplet against the source text through majority voting, filtering out hallucinated and structurally invalid assertions. A Contextual Profiler then enriches the surviving triples with semantic descriptors by situating each mention within its textual context, resolving polysemous cases such as distinguishing "Apple" as a company from "Apple" as a fruit.

For structural inconsistency (**Challenge 2**), another group of agents operates within a Structural Integration Layer. A Knowledge Linker compares semantic profiles of entities and relations to identify and merge redundant nodes, for example aligning "Barack Obama" and "the 44th U.S. President" into a single entity. An Ontological Logic Auditor then applies logical constraints to the graph, enabling it to both prune contradictory assertions and perform logical inference to refine the graph structure. Through this multi-agent collaboration, AgentsKG transforms unstructured text into a coherent knowledge graph without additional training.

The main contributions of this work are summarized as follows:

- **Multi-agent Framework:** We propose AgentsKG, a multi-agent framework for training-free Knowledge Graph Construction that explicitly decouples local semantic auditing from global structural integration. This multi-agent architecture enables specialized roles to resolve semantic and structural Challenges.
- **Open-domain Robustness:** We show that AgentsKG is effective in challenging open-domain and schema-free settings. The framework can successfully align diverse entities and relations and maintain consistency without relying on predefined schemas or prior domain knowledge.
- **Empirical Performance:** Extensive experiments on benchmark datasets, including WebNLG, CaRB, and AIDA-CoNLL, show that AgentsKG significantly outperforms state-of-the-art training-free baselines. Our approach demonstrates superior performance in both extraction reliability and the structural quality of the resulting knowledge graphs.

2 Related Work

2.1 LLMs and Autonomous Agents

The rapid evolution of LLMs has fundamentally transformed the landscape of natural language understanding and reasoning. Foundational models, such as GPT-4 [1] and Llama [31], exhibit remarkable in-context learning capabilities, allowing them to follow complex instructions without task-specific fine-tuning. Building on these foundations, the paradigm has shifted from single-pass generation toward autonomous agentic workflows. Key advances such as Chain-of-Thought (CoT) prompting [36], as well as its extensions that integrate dynamic graph structures to guide multi-hop reasoning paths [24], and the ReAct framework [43] enable LLMs to interleave reasoning traces with task-specific actions, facilitating multi-step problem solving. Furthermore, the emergence of multi-agent systems has introduced a collaborative dimension to LLM utilization. Frameworks like AutoGen [38] and Camel [21] demonstrate that complex tasks can be decomposed into specialized roles, such as planners, executors, and critics, which interact through consensus-seeking or debate to mitigate individual hallucinations and improve logical consistency [6]. Agents powered by LLM have demonstrated significant potential in various downstream applications, ranging from interactive decision-making to personalized recommendation systems [28]. These developments in autonomous planning, self-reflection, and collaborative reasoning provide a robust technological basis for tackling high-level knowledge modeling tasks that require both semantic and structural ability.

2.2 Knowledge Graph Construction

2.2.1 Traditional and Neural Paradigms. The KGC has evolved from rule-based pipelines to neural architectures. Early efforts, such as Stanford OIE [2], utilized hand-crafted linguistic patterns, but often yielded fragmented results. To improve precision, discriminative models such as OpenIE6 [19] integrated PLM backbones for sequence labeling, alongside various architectures for triple classification [33, 42]. Subsequently, the paradigm shifted towards generative Sequence-to-Sequence modeling, exemplified by REBEL [16], which treats relation extraction as a translation task [29, 41]. Despite their successes, these supervised paradigms remain heavily dependent on intensive labeling and fine-tuned static weights. Their restricted receptive fields and lack of adaptability limit their generalizability in open-domain scenarios, where capturing long-range dependencies and evolving schemas is essential.

2.2.2 LLM Paradigms. The ascent of LLMs has shifted KGC toward unified instruction-following paradigms. Frameworks such as InstructUIE [34] and Instruct-KGC [9] consolidate disparate extraction tasks into a single generative framework through supervised fine-tuning of task-specific instructions. This paradigm has been further specialized for vertical domains, with models such as BioMedGPT [25] and Pixiu [40] embedding domain-specific constraints directly into model parameters. However, even with parameter-efficient techniques like LoRA [15], these supervised architectures remain parameter-bound. Because knowledge is coupled with static weights, they lack the flexibility to incorporate emerging entities or evolving schemas in real-time. This dependency on costly retraining hinders scalability in high-velocity open-domain environments, necessitating a transition toward more agile, training-free frameworks. Additionally, beyond standard triple extraction, recent works have explored leveraging LLMs for structured knowledge modeling tasks, such as lightweight ontology learning [23]. Parallel to these supervised efforts, training-free paradigms leverage In-Context Learning to handle unconstrained environments without model updates [12]. A representative framework is SF-GPT [30], which operates in a schema-free manner by employing a lexical-centric alignment mechanism. Specifically, SF-GPT prompts the LLM to generate comprehensive alias sets for entities and performs merging by comparing the string distance between these sets. While effective at the lexical level, these frameworks often neglect relational consistency and lack rigorous quality control for entity merging.

A new paradigm leverages multi-agent collaboration to decompose KGC into specialized sub-tasks. Gui et al. [4] proposed a robust framework where specialized agents—including Knowledge Graph Experts and Extraction Experts—interact through a revision mechanism to refine preliminary graph structures. Despite these advances, existing approaches largely depend on implicit reasoning within LLMs and do not explicitly integrate entity and relation disambiguation with schema-level constraints. In this context, our work builds upon prior multi-agent KGC frameworks by introducing a Hierarchical framework. Without introducing additional supervision or model updates, our approach focuses on enforcing semantic perception and structural consistency during graph construction, aiming to improve the coherence and reliability of the resulting knowledge graph in complex open-domain scenarios.

3 Method

In this section, we present the architectural design of AgentsKG, a multi-agent collaborative framework for autonomous knowledge graph construction. The system is strategically organized into a two-layer hierarchy, where a collective of specialized agents $\mathcal{A} = \{\mathcal{A}_{\text{ext}}, \mathcal{A}_{\text{ver}}, \mathcal{A}_{\text{prof}}, \mathcal{A}_{\text{link}}, \mathcal{A}_{\text{ont}}\}$ collaborate to manage the transition from raw text to a logically consistent knowledge graph.

We formalize Open-Domain KGC not as a sequence-to-sequence generation, but as a hierarchical multi-objective optimization problem. Specifically, we argue that KGC comprises three sub-tasks with naturally conflicting objective functions: (1) Information Extraction, which prioritizes high recall to capture potential facts; (2) Semantic Alignment, which demands high precision to resolve entity and relation ambiguities; and (3) Ontological Modeling, which enforces global structural consistency.

In conventional end-to-end or single-pass prompting paradigms, LLMs are forced to balance these competing objectives implicitly within a single generative context. This often leads to Task Collapse, where the model sacrifices structural integrity for local fluency, resulting in factual noise or fragmented triple islands that fail to form a coherent graph. We posit that specialization does not imply unnecessary complexity, but is a necessary prerequisite for robustness. By explicitly decoupling these conflicting goals into a hierarchical multi-agent workflow, AgentsKG ensures that each agent operates within a focused optimization space, effectively transforming KGC from a fragile black-box generation into a transparent and auditable curation process. To ensure reproducibility and transparency of agentic behaviors, all prompt templates are collectively detailed in the appendix B.

3.1 Semantic Perception Layer

The Semantic Perception Layer serves as the gateway of AgentsKG, responsible for extracting candidate triples and refining their semantic attributes. This layer is composed of three primary agents: the Extraction Agent ($\mathcal{A}_{\text{ext}}$), the Verification Committee ($\mathcal{A}_{\text{ver}}$) and the Contextual Profiler ($\mathcal{A}_{\text{prof}}$).

A core design principle of our framework is the explicit decoupling of semantic roles to manage the divergent optimization pressures inherent in open-domain KGC. Conflating these objectives within a single prompt forces the model to be simultaneously expansive (for high-recall extraction via $\mathcal{A}_{\text{ext}}$) and restrictive (for high-precision verification via $\mathcal{A}_{\text{ver}}$), which often leads to unstable or hallucinated output. By separating these concerns, $\mathcal{A}_{\text{ext}}$ focuses purely on harvesting potential facts, $\mathcal{A}_{\text{ver}}$ filters structurally invalid or ungrounded assertions, and $\mathcal{A}_{\text{prof}}$ refines the semantics of surviving mentions via context-aware abstraction. This isolation ensures that each agent operates in a focused optimization space, producing clean, purpose-aligned inputs for downstream structural integration.

3.1.1 Triplet Extraction via $\mathcal{A}_{ext}$. The Extraction Agent functions as the initial knowledge harvester. Given a raw document $\mathcal{D}$, $\mathcal{A}_{\text{ext}}$ uses an LLM-based zero-shot (or few-shot) prompting strategy to identify all potential relational assertions. To ensure that the output is suitable for a formal KG, we impose strict constraints on entity atomicity and predicate meaningfulness. Instead of simple span clipping, $\mathcal{A}_{\text{ext}}$ is instructed to rephrase complex phrases into

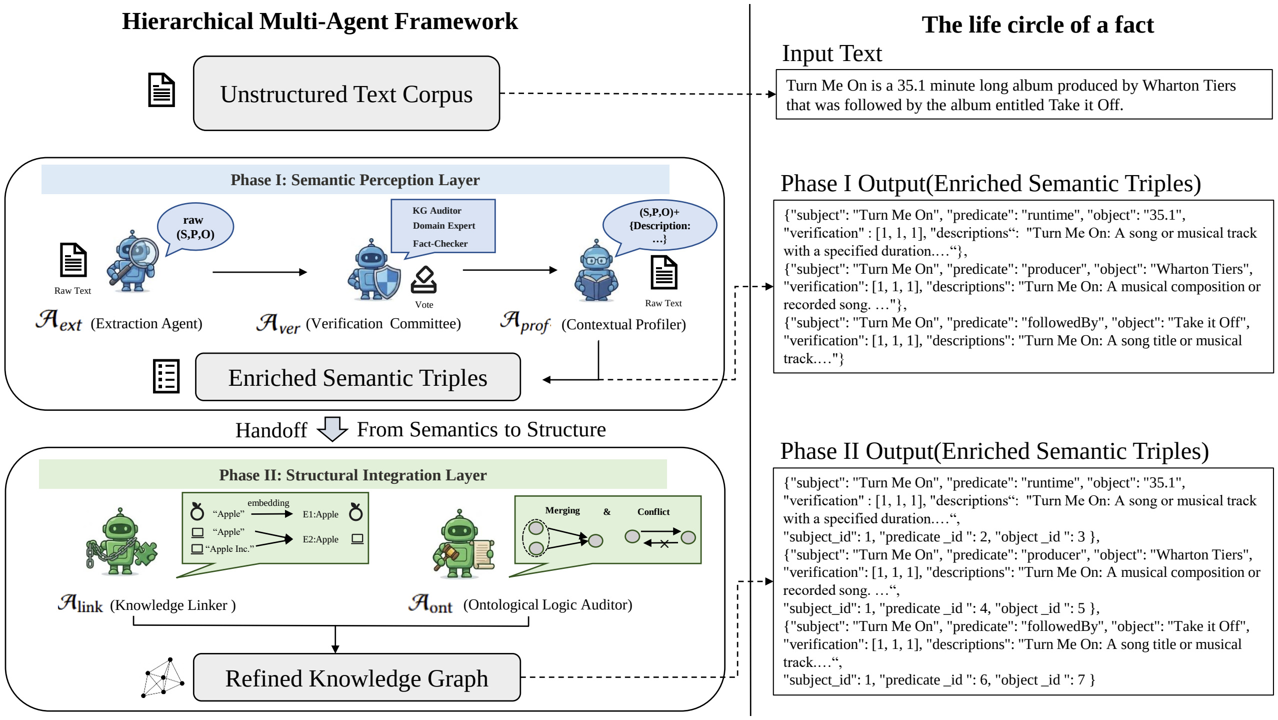

Figure 1: The hierarchical architecture of AgentsKG, illustrating the explicit decoupling of semantic perception from structural integration. The framework operates in two stages: (1) the Semantic Perception Layer, which leverages a collaborative verification committee and contextual profiling to produce enriched semantic triples; and (2) the Structural Integration Layer, which performs profile-driven alignment and deductive logic auditing to ensure global consistency. The right-hand panel traces the "life cycle of a fact" from raw unstructured text to its final canonicalized representation.

concise property names, yielding a candidate set $\mathcal{T}_{\text{raw}} = \{(s, p, o)_i\}$, where s, p, and o denote the subject, predicate and object of the i-th extracted assertion, respectively.

3.1.2 Consensus-based Auditing via $\mathcal{A}_{ver}$. To ensure the multi-dimensional integrity of the extracted knowledge, the candidate triples $\mathcal{T}_{\text{raw}}$ are subjected to a rigorous peer-review process by the Verification Committee ($\mathcal{A}_{\text{ver}}$).

Distinct from traditional self-consistency methods that rely on repeated sampling from a uniform perspective, our verification module is designed as a multi-role committee. This role-playing paradigm simulates a structured cross-examination process, where each agent is tasked with mitigating a specific category of extraction noise from an orthogonal viewpoint:

- **KG Auditor** enforces structural canonicalization. Ensures that each triplet is suitable for formal graph representation by strictly requiring atomic entities and avoiding descriptive phrases. As a result, every extracted element corresponds to a discrete, linkable concept rather than an unstructured text span.
- **Domain Expert** serves as a guardian of semantic plausibility. It evaluates whether the conceptual relationships are logically coherent within the domain context, filtering out assertions that may be grammatically correct but violate real-world knowledge or domain constraints.
- **Fact-checker** verifies factual fidelity. It cross-references each candidate (s, p, o) against the source document D to ensure that the assertion is explicitly supported by textual evidence, thus suppressing hallucinated facts.

Each triplet $t \in \mathcal{T}_{\text{raw}}$ receives a binary vote $v \in \{0, 1\}$ from each committee role. A triplet is promoted to the verified set $\mathcal{T}_{\text{ver}}$ if and only if it achieves a majority consensus (typically $\geq 2/3$). This consensus mechanism provides a lightweight approximation of ensemble verification without introducing additional supervision, ensuring that only robust and well-supported semantic units proceed to the structural integration stage.

3.1.3 Contextual Profiling via $\mathcal{A}_{prof}$. $\mathcal{A}_{\text{prof}}$ focuses on semantic enrichment. At this stage, triples consist of surface-level strings extracted directly from the text. To enhance these triples for further processing, $\mathcal{A}_{\text{prof}}$ generates a Semantic Profile for the subject, predicate, and object of each verified triplet. The agent's objective is to provide the semantic evidence required for downstream tasks. By re-examining the original document D, $\mathcal{A}_{\text{prof}}$, identifies the underlying ontological category of each term and formulates a concise, class-level definition—for instance, defining "Apple" as a "technology corporation" rather than a "type of fruit", based on its usage in the text. These profiles act as conceptual anchors, transforming simple strings into well-defined entities. This enrichment is a prerequisite for the next layer, as it provides the descriptive features necessary for the Knowledge Linker to perform accurate entity resolution and relation merging.

3.2 Structural Integration Layer

The Structural Integration Layer focuses on organizing verified semantic fragments into a globally consistent knowledge graph. While the Semantic Perception Layer emphasizes factual correctness at the instance level, this stage addresses structural alignment decisions, including entity identity resolution and relation canonicalization. These decisions determine whether newly extracted semantic units should be merged with existing graph components or instantiated as distinct nodes and relations.

In open-domain settings, such alignment decisions are often under-specified by surface forms or embedding similarity alone. Pure similarity-based matching may lead to erroneous merges, while exhaustive reasoning over all candidates is computationally infeasible. Accordingly, structural integration in AgentsKG is formulated as a selective decision process: lightweight similarity signals are first used to narrow down plausible alignment candidates, and explicit agent reasoning is invoked only when necessary.

3.2.1 Structural Integration and Merging via $\mathcal{A}_{link}$. The Knowledge Linker ($\mathcal{A}_{\text{link}}$) serves as the central agent responsible for structural integration. Its primary role is to consolidate fragmented assertions into a coherent graph structure by resolving entity identity and relation equivalence.

Each semantic unit—entity or relation—is represented by a profile constructed in the Semantic Perception Layer. $\mathcal{A}_{\text{link}}$ encodes these profiles using the BGE-M3 embedding model, producing dense vectors $\mathbf{v} \in \mathbb{R}^{1024}$. These embeddings are not used to make final alignment decisions directly, but to estimate semantic proximity and to identify cases where merging decisions are straightforward versus cases requiring explicit verification.

To operationalize this process, $\mathcal{A}_{\text{link}}$ partitions the similarity space into three confidence regions based on cosine similarity $S = S_{\cos}(\mathbf{v}_i, \mathbf{v}_j)$ between candidate pairs (x_i, x_j):

(1) **Direct Identity Zone ($S \geq \tau_{high}$):** Pairs with very high similarity are treated as referring to the same semantic unit. These cases correspond to clear lexical or semantic variants (e.g., USA vs. United States) and are merged automatically without further agent involvement.
(2) **Agent Verification Zone ($\tau_{low} \leq S < \tau_{high}$):** In this intermediate region, similarity alone is insufficient to determine equivalence. For such cases, $\mathcal{A}_{\text{link}}$ triggers an explicit verification step, consulting the semantic profiles and, when needed, delegating the decision to the multi-agent committee. The committee returns a binary decision indicating whether the two units should be merged or kept separate.
(3) **Separation Zone ($S < \tau_{low}$):** Pairs with low similarity are considered semantically distinct and are instantiated as separate entities or relations without further inspection.

This confidence-guided integration strategy allows $\mathcal{A}_{\text{link}}$ to limit expensive reasoning operations to a small subset of structurally ambiguous cases, while handling the majority alignments through efficient similarity filtering.

Entity Resolution. For entities, $\mathcal{A}_{\text{link}}$ compares profile embeddings to resolve lexical variation and synonymy. Mentions with highly similar profiles are merged into a single node, increasing graph connectivity. Conversely, mentions sharing identical surface forms but differing in contextual semantics yield low similarity and are maintained as distinct nodes. This mechanism prevents unintended collapses while preserving the necessary granularity.

Relation Canonicalization. Relation phrases are processed analogously. By embedding relation profiles into the same semantic space, $\mathcal{A}_{\text{link}}$ identifies predicates that convey equivalent relational semantics. Lexically diverse expressions such as "was born in" and "place of birth" can thus be aligned to a shared canonical relation. This many-to-one mapping reduces relational redundancy and progressively regularizes the predicate space, bringing the open-domain graph closer to a schema-consistent structure.

Through unified handling of entity and relation alignment, $\mathcal{A}_{\text{link}}$ ensures that semantically redundant subgraphs are consolidated into a single representation. The resulting structure provides a stable and compact foundation for subsequent ontology auditing and global consistency checks.

3.2.2 Ontological Logic Auditing via $\mathcal{A}_{ont}$. Guided by recent algebraic foundations for Knowledge Graph construction [26], the Ontological Logic Auditor ($\mathcal{A}_{\text{ont}}$) serves as the symbolic reasoning engine of AgentsKG. It ensures that the constructed graph $\mathcal{G}$ is not simply a collection of mentions, but a logically consistent knowledge system. Unlike extractors that treat relations as surface-level labels, $\mathcal{A}_{\text{ont}}$ conceptualizes each relation p as a logical predicate governed by a 7-dimensional property vector $\mathbf{v}_{\mathcal{P}} \in \{0, 1\}^7$.

Pragmatic Property Inference. To handle the inherent noise of open-domain extraction, $\mathcal{A}_{\text{ont}}$ adopts a pragmatic relaxed constraints strategy. The complete catalog of logical dimensions and their formal definitions are provided in Appendix A.

Crucially, the agent is instructed to prioritize data retention: a relation is only marked as Functional (1) if it is strictly limited to exactly one value (e.g., hasBirthDate). For ambiguous relations such as hasChild or spouse (considering remarriage histories), the auditor defaults to a non-functional setting (0). This ensures that the schema remains robust against real-world complexity while maintaining rigor where absolute laws apply.

Operational Refinement. Based on the inferred vector $\mathbf{v}_{\mathcal{P}}$, $\mathcal{A}_{\text{ont}}$ executes two critical graph-optimization operations:

- Conflict Elimination: The agent prunes "logical hallucinations." For example, if p is *Asymmetric*, the auditor removes bidirectional contradictions $\{(A, p, B), (B, p, A)\}$ based on confidence scores.
- Deductive Augmentation: For relations identified as *Transitive* (e.g., *isLocatedIn*), $\mathcal{A}_{\text{ont}}$ performs a logical closure operation. It infers implicit triples to bridge structural gaps, transforming fragmented assertions into a dense, interconnected knowledge system.

4 Experiments

In this section, we present a comprehensive experimental evaluation of AgentsKG. We evaluate the framework's performance across diverse benchmarks to demonstrate its efficacy in handling complex relations, open-domain extraction, and semantic entity alignment.

4.1 Experimental Setup

4.1.1 Datasets. We conduct experiments on three datasets, each targeting a specific facet of the KGC task: **WebNLG** [47]: A multi-domain benchmark repurposed for triple extraction across 15 semantic categories (e.g., Building, Sports). It is used here to evaluate AgentsKG's ability to handle complex multi-triplet structures and diverse relational predicates within intricate syntactic environments. **CaRB** [3]: A gold-standard for Open Information Extraction. It provides a rigorous testbed for schema-free extraction in heterogeneous, unconstrained scenarios, allowing us to assess the robustness of our framework in the absence of predefined ontologies. **AIDA-CoNLL** [13]: A data set derived from Reuters news articles, widely used for Named Entity Disambiguation. Given its high density of polysemous mentions, it is utilized to assess the proficiency of our Semantic Perception Layer in mention-level entity identification and contextual resolution.

4.1.2 Baselines. We compare AgentsKG with representative methods from four categories:

- **Traditional & Neural OIE**: Stanford OIE [2], a rule-based system that utilizes Natural Logic and clause simplification for pattern-based extraction; and OpenIE6 [19], which adopts an Iterative Grid Labeling paradigm to extract triples in a BERT-based 2D coordinate space.
- **Strategy-specific LLM Frameworks**: (1) **DualOIE** [5], which decomposes the extraction task into a dual-stage cognitive process via Chain-of-Thought prompted to reduce reasoning complexity. (2) **PiVe** [10], which introduces an iterative self-correction loop within an extraction-then-verification paradigm to refine factual accuracy and mitigate hallucinations. (3) **SF-GPT** [30], a schema-free pipeline that achieves disambiguation by evaluating the string distance between LLM-generated entity alias sets. Althrough the original work suggests a threshold of $s = 0.6$, we empirically observed suboptimal results and adjusted it to $s = 0.7$ to ensure a more competitive baseline.
- **Foundation & Reasoning LLMs**: To establish the performance floor and evaluate raw capabilities, we include: (1) Llama-3-8B-Instruct, (2) DeepSeek-R1-8B, and (3) Qwen3-30B-A3B-Instruct. These models are evaluated in zero-shot and few-shot settings.

4.1.3 Evaluation Metrics. To rigorously assess the performance of **AgentsKG** under various lexical and structural constraints, we adopt the following metric protocols: In WebNLG, we follow the standard WebNLG evaluation protocol for triple-level comparison. The predicted triples are evaluated against the gold triples using exact matching in the subject, relation, and object fields. In particular, underscores are treated as spaces following the WebNLG convention, while parenthesized type hints present in some gold entities (e.g., (album)) are not explicitly handled, as entity types are already distinguished in our system. This setup ensures consistency with prior WebNLG-based KGC evaluations.

In CaRB, we adopt the official Default Match evaluation, which computes token-level F1 over binarized n-ary tuples. This metric balances linguistic flexibility with structural correctness and is the standard protocol for Open IE evaluation.

In AIDA-CoNLL, the identification of an entity is evaluated using mention-level Precision, Recall, and F1. Predictions are considered correct if they exactly match the gold mention or satisfy a relaxed substring inclusion criterion after normalization. This protocol accommodates minor surface variation while preserving mention-level accuracy.

4.1.4 Implementation Details. We implement AgentsKG by interfacing with the official APIs for Qwen2.5-7B-Instruct and Qwen3-30B-A3B-Instruct. These two models of varying scales are employed independently to evaluate the framework's adaptability across different parameter capacities. To ensure a fair comparison, all other baseline frameworks are implemented using Qwen3-30B-A3B-Instruct as their common backbone. To ensure reproducibility and minimize variance, all requests are executed with the temperature set to 0. Based on the sensitivity analysis in the hyperparameter study, we set the threshold to $\tau_{high} = 0.90$ and $\tau_{low} = 0.80$. To accommodate the diverse structural requirements of each benchmark, we utilize task-aware prompts. In a few-shot settings, we set $k = 3$ using in-domain demonstrations. Crucially, while prompts are adapted to specific task contexts, the underlying hierarchical agentic architecture and ontological auditing logic remain strictly identical across all experiments, ensuring a consistent and autonomous construction pipeline.

4.2 Main Results

Table 1 summarizes the performance of all methods on WebNLG, CaRB, and AIDA-CoNLL. Across all benchmarks, AgentsKG consistently achieves state-of-the-art results, outperforming both task-specific pipelines and strong general-purpose LLMs. These results demonstrate the effectiveness of our hierarchical multi-agent framework in addressing diverse challenges in open-domain knowledge graph construction.

On the structurally demanding WebNLG benchmark, AgentsKG-30B reaches the highest F1 score (35.5%). The overall low absolute scores across all methods reflect the strict evaluation protocol, which requires exact schema-level matching for abstractive relations. In this setting, AgentsKG shows a clear advantage in aligning semantically equivalent but lexically diverse relational expressions, leading to substantial gains over all baselines.

In the AIDA-CoNLL data set, AgentsKG-30B (zero-shot) achieves a new state-of-the-art F1 score of 58.8%. This result indicates that AgentsKG is more effective in identifying and anchoring entity mentions in dense text, even under flexible surface forms.

In CaRB, AgentsKG-30B (few-shots) achieves an F1 score of 52.5%, substantially outperforming all baselines. Although vanilla LLMs tend to favor high recall at the cost of precision in unconstrained extraction, AgentsKG maintains a significantly better precision–recall balance. This indicates that the framework effectively suppresses spurious extractions while preserving broad coverage, which is crucial for open information extraction.

Table 1: Performance comparison on three benchmarks (%). We evaluate AgentsKG with two backbones (Qwen2.5-7B, Qwen3-30B) under zero-shot and few-shot settings. The best results are bold, and second best are underlined.

Model	WebNLG			AIDA-CoNLL			CaRB		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
Stanford OIE	0.0	0.0	0.0	29.7	69.2	41.5	19.5	24.9	21.9
OpenIE6	-	-	-	-	-	-	52.9	41.7	46.6
PiVe	1.7	1.7	1.7	39.0	89.3	54.2	52.6	35.6	42.5
DualOIE	0.2	0.3	0.2	38.7	58.8	46.7	43.8	41.7	42.7
SF-GPT ($s = 0.7$)	13.3	19.0	15.7	42.2	27.9	33.6	36.8	35.6	36.2
Llama-3-8B (Zero)	1.1	1.3	1.2	40.0	64.2	49.3	34.8	30.6	32.6
Llama-3-8B (Few)	21.8	22.4	22.1	42.1	74.6	53.9	40.9	35.9	38.2
DeepSeek-R1-8B (Zero)	2.0	1.7	1.8	<u>49.2</u>	40.9	44.6	47.7	21.8	29.9
DeepSeek-R1-8B (Few)	15.1	13.3	14.1	44.2	39.8	41.9	55.2	31.6	40.2
Qwen3-30B (Zero)	1.5	1.6	1.6	38.4	89.2	53.7	48.3	35.7	41.1
Qwen3-30B (Few)	2.2	2.3	2.3	41.1	**96.4**	<u>57.6</u>	58.3	<u>42.3</u>	<u>49.1</u>
AgentsKG-7B (Zero)	30.6	25.6	27.9	45.5	35.8	40.0	61.0	30.9	41.0
AgentsKG-7B (Few)	34.8	28.8	31.5	**49.7**	35.6	41.5	**65.7**	39.1	49.0
AgentsKG-30B (Zero)	**35.8**	<u>34.9</u>	<u>35.3</u>	44.2	88.0	**58.8**	56.5	38.2	45.6
AgentsKG-30B (Few)	<u>35.5</u>	**35.6**	**35.5**	41.6	<u>91.7</u>	57.3	<u>65.2</u>	**44.0**	**52.5**

4.3 Ablation Study

In this section, we conduct an ablation study on the AIDA-CoNLL data set to analyze how individual modules contribute to entity canonicalization and graph coherence. To disentangle the extraction quality from the graph construction quality, we adopt a two-tier evaluation protocol. First, we report standard mention-level metrics (Precision, Recall, and F1), which reflect the system's ability to identify entity mentions from text. However, accurate mention extraction alone does not guaranty a coherent knowledge graph. To assess whether extracted mentions are correctly canonicalized into unique entity nodes, we further introduce entity-level alignment metrics:

- **Coverage (C):** Measures the proportion of gold entities that are recovered, defined as $C = \frac{|E_{ext} \cap E_{gold}|}{|E_{gold}|}$.
- **Purity (P):** Evaluates the quality disambiguation by computing the weighted dominance of a single gold entity within each predicted cluster ω_k: $P = \frac{1}{N} \sum_k \max_j |\omega_k \cap c_j|$, where c_j denotes the IDs of the gold entity and N is the total number of mentions.
- **Homogeneity (H):** Assesses aggregation consistency based on conditional entropy, reaching its maximum when all mentions of a gold entity are assigned to a single predicted cluster.

Limitations of Mention-Level Metrics. The mention-level F1 scores in Table 2 remain relatively stable in all variants (57.3%–58.9%), indicating comparable surface extraction capability. However, this apparent stability masks substantial differences in graph quality. Variants without the Contextual Profiler ($\mathcal{A}_{\text{prof}}$) suffer a severe collapse in Purity (down to 1.0%), revealing that the resulting graph degenerates into fragmented and ambiguous entity strings despite correct mention extraction.

Effect of Semantic Anchoring and Linking. Entity-level diagnostics highlight the necessity of the AgentsKG design. Removing $\mathcal{A}_{\text{prof}}$ prevents the system from producing the semantic representations required to support confidence-based aggregation, rendering the Knowledge Linker ineffective in resolving ambiguous mentions. As a result, entity clusters lose semantic consistency. Similarly, removing the Knowledge Linker ($\mathcal{A}_{\text{link}}$) leads to a sharp drop in Homogeneity (56.6% $\longrightarrow$ 18.2%), indicating that coherent entity aggregation relies on explicit linking mechanisms rather than surface-level LLM generation alone.

Effect of Ontological Logic Auditing. To evaluate the impact of $\mathcal{A}_{\text{ont}}$ on graph-level consistency, we quantify its trigger frequency in various logical properties in Table 3. Our results demonstrate that $\mathcal{A}_{\text{ont}}$ is instrumental in preserving the logical integrity of the knowledge graph. The module intercepted 143 functional violations and resolved 226 deduplication errors, which would otherwise lead to semantic collapse—where conflicting facts about the same entity coexist in the graph.

Although the triggering frequency for complex properties,such as transitivity, is relatively low, this is a design choice to prioritize high-fidelity representation. For example, in a case study of spatial and hierarchical relations, the module successfully applied transitive reasoning to 3/7 triples involving the [are part of] relation and 8/151 triples for [located in].

4.4 Analysis

To evaluate the knowledge generated beyond the primary F1 scores, we analyze its **extraction robustness** and **relational consistency**.

Robustness: Schema Mismatch vs. Extraction Failure. The WebNLG benchmark's strict ontology often penalizes semantically valid but free-form predicates (e.g., born in vs. birthPlace). To decouple the factual extraction capability from schema alignment, we perform an Entity-Pair Matching (Relaxed Match) analysis.

As shown in Table 4, all models exhibit a universal performance surge under this protocol. Notably, AgentsKG-30B's F1 jumps to **57.9%**, indicating that the majority of its "errors" under strict evaluation are actually valid predicates mismatched with the specific schema. Although DeepSeek-R1 shows conservative extraction (only 33.7% recall), AgentsKG-30B maintains superior recall (57.7%), reflecting its ability to capture diverse factual associations. Crucially, our 7B variant outperforms the 30B Qwen3 baseline (50.7% vs.

Table 2: Ablation study on AIDA-CoNLL. Metrics include mention-level performance (P, R, F1) and entity-alignment diagnostics (Coverage, Purity, Homogeneity). $w/o\ \mathcal{A}_{\text{ver}}$ & $\mathcal{A}_{\text{prof}}$ denotes the removal of both verification and profiling components.

Model Variant	P	R	F1	Triples	Coverage	Purity	Homog.	Ent. F1
AgentsKG (Full)	41.6	91.7	57.3	6,568	89.6	87.4	56.6	**68.7**
$w/o\ \mathcal{A}_{\text{ver}}$ (Verification)	42.2	96.9	58.8	6,677	91.8	86.9	54.7	67.1
$w/o\ \mathcal{A}_{\text{link}}$ (Linker)	41.6	91.7	57.3	6,568	89.6	89.1	18.2	30.2
$w/o\ \mathcal{A}_{\text{prof}}$ (Profiler)	42.6	95.4	58.9	6,568	91.8	1.0	100.0	1.9
$w/o\ \mathcal{A}_{\text{ver}}$ & $\mathcal{A}_{\text{prof}}$	42.7	95.2	58.9	6,844	91.5	1.0	100.0	2.0

Table 3: Statistics of $\mathcal{A}_{\text{ont}}$ logic auditing on subset of AIDA-CoNLL. "Checks" denotes the total evaluations performed, while "Violations" and "Actions" represent the number of intercepted conflicts and triggered completion actions, respectively.

Logic Property	Checks	Violations	Actions
Functional	883	143	-
Inverse-Functional	203	42	-
Transitive	401	-	11
Symmetric	-	-	248
Asymmetric	3,184	13	-
Reflexive	-	-	60
Irreflexive	4,348	5	-
Deduplication	5,885	226	-

Table 4: Entity-pair matching (Relaxed Match) results on WebNLG. This evaluation focuses on the model's ability to capture core factual associations by decoupling subject-object identification from strict relation schema constraints. The best results are bold, and second best are underlined.

Category	Model	P	R	F1-Soft
External Baselines	Stanford OIE	5.7	9.0	7.0
	DualOIE	27.5	35.4	31.0
	PiVe	45.1	47.3	46.2
	SF-GPT	43.9	52.0	47.6
Individual LLMs	Llama-3-8B (Zero)	29.9	33.0	31.4
	Llama-3-8B (Few)	41.1	42.2	41.6
	DeepSeek-R1-8B (Zero)	39.8	33.7	36.5
	DeepSeek-R1-8B (Few)	44.7	40.1	42.3
	Qwen3-30B (Zero)	44.0	46.8	45.4
	Qwen3-30B (Few)	44.2	45.7	44.9
Ours	**AgentsKG-7B (Zero)**	55.1	47.0	50.7
	AgentsKG-7B (Few)	**59.6**	50.3	54.6
	AgentsKG-30B (Zero)	58.1	<u>57.7</u>	<u>57.9</u>
	AgentsKG-30B (Few)	<u>58.8</u>	**59.8**	**59.3**

45.4%), proving that our hierarchical agentic architecture is more decisive than the scale of the raw parameter in complex extraction tasks.

Structural integrity depends on a normalized predicate space. We analyze the consolidation of the relation on CaRB using two macroscopic metrics in the set of relation R:

Table 5: Comparative analysis of relationship consistency. $Den > \alpha$ represents the redundancy density at similarity threshold α. ANRS denotes the Average Normalized Redundancy Score (lower is better).

Model / Method	Triples	Unique Rel.	ANRS ↓	Redundancy Density ($Den > \alpha$)		
				> 0.9	> 0.8	> 0.7
Stanford OIE	5,776	1,202	0.8387	20.050	86.938	393.261
OpenIE6	1,883	1,105	0.8133	9.774	78.552	830.950
Qwen3-30B (Zero)	784	679	0.7044	0.295	3.976	155.965
Qwen3-30B (Few)	1,429	871	0.7545	0.230	14.237	955.683
Qwen2.5-7B (Zero)	1,054	692	0.7579	0.723	13.584	715.751
Qwen2.5-7B (Few)	1,139	790	0.7494	0.253	9.367	620.000
SF-GPT ($s = 0.7$)	5,821	2,808	0.8545	23.590	122.650	955.874
PiVe	2,513	1,557	0.8179	11.111	82.787	708.221
AgentsKG-7B (Zero)	4,120	1,850	0.7105	0.210	3.450	55.200
AgentsKG-7B (Few)	4,890	2,100	0.7012	0.190	3.100	50.150
AgentsKG-30B (Zero)	6,210	2,350	0.6920	0.165	2.540	48.300
AgentsKG-30B (Few)	6,568	2,410	0.6850	0.150	2.120	45.600

- **ANRS** (Average Nearest Relationship Similarity) ↓: Measures semantic separability. Lower values indicate clearer boundaries and less semantic collapse.

$$\text{ANRS}(R) = \frac{1}{|R|} \sum_{i=1}^{|R|} \max_{j \neq i} \text{sim}(\mathbf{v}_{r_i}, \mathbf{v}_{r_j}),$$

  where R is the set of relations, $|R|$ denotes its cardinality, and $\mathbf{v}_{r_i}$ represents the embedding vector of relation r_i. The function $\text{sim}(\cdot, \cdot)$ measures the cosine similarity between two vectors.
- **Den$_{>x}$** (Normalized Redundancy Density) ↓: Reports the number of near-duplicate relation pairs (similarity $> x$) per 100 unique relations.

Observations on Relational Quality. As summarized in Table 5, AgentsKG-30B achieves an ANRS of 0.6850, displaying a more discriminative predicate space than traditional OIE baselines. In terms of redundancy, AgentsKG maintains a density of 0.3–0.7 for $Den_{>0.9}$, whereas the alias-based SF-GPT suffers from a massive redundancy of 23.6. This confirms that our Knowledge Linker effectively consolidates equivalent predicates, ensuring a compact and functional graph structure.

Cost-Effectiveness and Performance Analysis. To address concerns regarding the computational overhead and the algorithmic necessity of our hierarchical framework, we conduct a comprehensive evaluation against a high-performance single-pass model (GPT-4o) and two representative schema-based multi-agent paradigms: (1) a stage-wise decomposition method ([4]), which extracts entities

Table 6: Cost-benefit analysis: performance and token costs

Method	Prec.	Rec.	F1	Coverage	Purity	Homog.	Tokens/Triple
GPT-4o (Single-pass)	0.692	0.391	0.500	0.435	0.960	1.000	300
Chen et al. [4]	0.520	0.565	0.541	0.783	0.707	0.285	200
Ye et al.[44]	0.688	0.164	0.265	0.236	0.966	0.689	5.5k
AgentsKG-30B (Ours)	0.652	0.440	0.573	0.896	0.874	0.566	3.4k

before relations; and (2) a multi-turn feedback method ([44]), which relies on iterative agent dialogues. The results are summarized in Table 6.

Recall Collapse in Single-pass Paradigms. As shown, while GPT-4o achieves high precision, it suffers from a low Recall. Our analysis reveals that single-pass models are fundamentally constrained by the context window when processing dense relational text, leading them to overlook critical long-tail entities (e.g., *FIFA*, *United Arab Emirates*). This demonstrates that even powerful closed-source models cannot maintain a high extraction density and global consistency simultaneously without an explicit decoupling of perception and integration.

The Bottleneck of Other Multi-Agents. The schema-driven approach exhibits a failure in Homogeneity, indicating that it is insufficient for resolving the polysemy inherent in open domains.

More notably, the multi-turn feedback framework [44] incurs the highest cost while yielding the lowest Recall. The accumulation of dialogue history in iterative interactions rapidly consumes the LLM's context window. To prevent overflow, aggressive truncation is required (as suggested by the original implementation), resulting in the catastrophic loss of information.

Industrial Feasibility and Cost Optimization. From an architectural standpoint, AgentsKG is designed to be model-agnostic. By delivering superior F1-scores using **30B-parameter open-source models**, our framework supports local deployment on private infrastructure, effectively eliminating the prohibitive financial costs of proprietary APIs. Furthermore, since the roles within our framework are functionally decoupled, modules focused on discriminative judgment rather than creative generation—such as $\mathcal{A}_{\text{ver}}$ (verification) and $\mathcal{A}_{\text{link}}$ (linking)—can be offloaded to even smaller models (e.g., 8B parameters or task-specific SLMs). It allows for further multi-fold reductions in computational overhead, providing a cost-effective solution for large-scale industrial KGC.

4.5 Hyperparameter Study

To investigate the impact of decision boundaries on entity resolution and computational efficiency, we conducted a grid search over the **high-confidence threshold** $\tau_{\text{high}} \in \{0.85, 0.90, 0.95\}$ and the **low-confidence threshold** $\tau_{\text{low}} \in \{0.4, 0.6, 0.8\}$. As summarized in Table 7, the results reveal a clear trade-off between system performance and operational cost.

Impact of the Low-Confidence Threshold. The threshold τ_{low} serves as the primary regulator of computational cost. We observed a sharp decline in LLM calls as τ_{low} increased from 0.6 to 0.8, with total API requests dropping by approximately 25% (from $\sim 12,000$ to $\sim 9,000$). This occurs because a higher τ_{low} shrinks the *Agent Verification Zone*, thus reassigning a significant volume of candidate pairs with moderate similarity directly to the *Separation Zone.* Although

Table 7: Hyperparameter sensitivity analysis on AIDA-CoNLL. Metrics include Coverage, Purity, Homogeneity, and entity-F1 (%). $\tau_{\text{low}} = 0.8$ yields the best trade-off between precision and efficiency.

τ_{high}	τ_{low}	C	P	H	entity-F1	LLM Calls
0.85	0.4	89.2	85.9	65.2	74.1	12,158
	0.6	89.2	85.8	65.2	74.1	12,114
	0.8	89.2	87.6	61.8	72.4	9,080
0.90	0.4	89.2	85.6	65.3	74.1	12,206
	0.6	89.2	85.7	65.1	74.0	12,115
	0.8	89.2	87.7	61.9	72.6	9,082
0.95	0.4	89.2	85.2	65.0	73.8	12,144
	0.6	89.2	85.8	65.2	74.1	12,112
	0.8	89.2	87.6	61.9	72.5	9,077

a lower threshold ($\tau_{\text{low}} = 0.6$) yields a marginal 1.5% improvement in F1, it incurs a disproportionate 33% increase in LLM calls. Consequently, we identify $\tau_{\text{low}} = 0.8$ as the optimal boundary to maximize Purity ($P \approx 0.877$) while minimizing overhead.

Stability of the High-Confidence Boundary. The system demonstrates high robustness to variations in the high-confidence threshold τ_{high}. Across the range of $\{0.85, 0.90, 0.95\}$, the entity-F1 score remains remarkably stable with negligible fluctuations. This stability indicates that the committee effectively manages the uncertainty in the verification zone; even when the automatic merging criteria are tightened (higher τ_{high}), the specialized agents provide consistent resolution logic that preserves Coverage (C).

Default Configuration. Based on Pareto-optimal results, we select $\tau_{\text{high}} = 0.90$ and $\tau_{\text{low}} = 0.80$ as the default configuration for AgentsKG. This setting ensures balanced construction performance while maintaining a low computational footprint, making the framework suitable for large-scale knowledge graph construction.

5 Conclusion

In this paper, we introduced AgentsKG, a hierarchical and training-free multi-agent open-domain KGC framework. By decoupling the KGC process, AgentsKG can isolate semantic perception from structural integration. This coordinated workflow ensures that the resulting graph is not merely a collection of extracted facts, but a logically coherent and structurally dense representation of knowledge. Extensive evaluations demonstrate that AgentsKG outperforms state-of-the-art training-free baselines. By bridging the gap between raw linguistic extraction and formal knowledge modeling, this work establishes a robust, training-free foundation for open-domain KGC. Future research will explore extending AgentsKG to multi-modal data and autonomous, self-evolving knowledge environments in dynamic scenarios.

6 Acknowledgement

This work was supported by the National Natural Science Foundation of China (NO.62472305) and the Key R&D project in Hainan province,China (NO.ZDYF2024SHFZ051).

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A Formal Definitions of Relational Properties

In Section Three, we introduce the Ontological Logic Auditor ($\mathcal{A}_{\text{ont}}$). Here, we provide the formal definitions for the seven logical dimensions evaluated by the agent. Table 8 summarizes the audit logic.

Table 8: Formal definitions and auditing logic for the seven relational properties.

Property	Formal Definition
Functional	$\forall a, b, c : \langle a, p, b \rangle \wedge \langle a, p, c \rangle \Rightarrow b = c$
Inv-Functional	$\forall a, b, c : \langle a, p, c \rangle \wedge \langle b, p, c \rangle \Rightarrow a = b$
Transitive	$\forall a, b, c : \langle a, p, b \rangle \wedge \langle b, p, c \rangle \Rightarrow \langle a, p, c \rangle$
Symmetric	$\forall a, b : \langle a, p, b \rangle \Rightarrow \langle b, p, a \rangle$
Asymmetric	$\forall a, b : \langle a, p, b \rangle \Rightarrow \neg \langle b, p, a \rangle$
Reflexive	$\forall a \in \mathcal{E} : \langle a, p, a \rangle$
Irreflexive	$\forall a \in \mathcal{E} : \neg \langle a, p, a \rangle$

B Prompt Templates

We provide the exact prompt templates used in our experiments to ensure full reproducibility. Our templates utilize several dynamic placeholders: {system_role} are as follows; {rules} outlines task constraints; {examples} provides few-shot demonstrations; input variables (e.g., {input_text}, {triple}) import the target data; and {output_format} defines output structure.

The `{system_role}` placeholder is instantiated using one of the following personas:

- You are a Knowledge Graph Auditor. Verify that the triplet is structurally sound: entities must be atomic (no long sentences), and the predicate must be a concise, valid property.
- You are a Domain Expert. Verify that the technical terms and concepts are used accurately and make logical sense within the given context.
- You are a strict Fact-Checker. Verify that the triplet is explicitly supported by the provided source text and contains NO hallucinations.

The full template is provided in the listing below:

```
Extraction Prompt Template:
You are a precision Knowledge Graph extraction system. Your task is to extract structured knowledge
triples (subject, predicate, object) from the given text.
Rules of Extraction:{rules}
Here are examples of the required extraction style:{examples}(optional)
Here is the text to be processed:{input_text}
Output Format:{output_format}

Verification Prompt Template:
System role:{system role}
Evaluate each triplet based on your assigned System Role. You must output a single Python list of
integers (0 or 1) corresponding strictly to the order of the input triples.
-  1 = Pass / Valid / True
-  0 = Fail / Invalid / False
Examples:{examples}
Source Text:{input_text}
Input Triples:{triples}
Output Format:{output_format}

Description Prompt Template:
You are a Knowledge Engineer building a formal ontology.Your task is to define the fundamental class
or category of a given target entity/relation. I will provide a triple (subject-predicate-object) to give you
semantic context, and a target term which is one of the entities/relation from that triple.
Your mission is to generate a concise, accurate, and universally applicable one-sentence definition for the
target term itself.
Rules of Description:{rules}
Examples:{examples}
Input Triples:{triple}
Target:{entity/relation}

Canonicalization Prompt Template:
You are a meticulous Knowledge Graph Ontologist specializing in entity/relation resolution. Your task is
to determine if a 'Target Entity/Relation' is semantically identical to one of the 'Candidate
Entities/Relations' and can be merged.
Rules of Judgement:{rules}
Examples:{examples}
Target Entity/Relation: {entity/relation}
Description: {description}
Candidate Entities/Relations:{other_entities/relations}
Output Format:{output_format}

Ontological Logic Prompt Template:
You are a pragmatic Knowledge Graph Engineer. Your task is to analyze the logical properties of a
relation to determine database constraints. Logical Property Definitions: Evaluate the relation against
these seven definitions:{logic definition}
Rules of Judgement:{rules}
Examples:{examples}
Input Relation Description:{relation}
Output Format:{output_format}
```

Figure 2: prompt templates designed for agents.

C Detailed Algorithmic Orchestration and Implementation

To facilitate the reproducibility of our framework, we provide the formal pseudocode for the AgentsKG construction pipeline below.

Algorithm 1: AgentsKG Orchestration and Construction Pipeline

Input: Document stream $\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$; Similarity thresholds τ_{high}, τ_{low}
Output: Refined Global Knowledge Graph $\mathcal{G}$

```
Initialize Global KG 𝒢 ← ∅;
// Layer 1: Semantic Perception
foreach document d_i ∈ 𝒟 do
    𝒯_raw ← Agent_ext(d_i) ;                               // Extract candidate triplets
    𝒯_verified ← ∅;
    foreach triple t ∈ 𝒯_raw do
        // Parallel multi-role verification (2/3 Majority Voting)
        v_fact ← FactChecker(t, d_i);
        v_kg ← KGAuditor(t);
        v_log ← LogicExpert(t);
        if ∑(v_fact, v_kg, v_log) ≥ 2 then
            𝒯_verified ← 𝒯_verified ∪ {t};
        end
    end
    // Generate contextual profiles to enrich latent semantics
    𝒫 ← Agent_prof(𝒯_verified, d_i);
    // Layer 2: Structural Integration
    foreach t ∈ 𝒯_verified do
        foreach mention m ∈ {t.subj, t.obj, t.rel} do
            C_cand ← BGE_M3_Retrieval(m, 𝒫_m, 𝒢) ;          // Retrieve alignment candidates
            if max(Sim(m, C_cand)) > τ_high then
                Merge m into highest similarity anchor in 𝒢;
            else if max(Sim(m, C_cand)) > τ_low then
                anchor ← Agent_link(m, 𝒫_m, C_cand) ;      // LLM disambiguation
                Merge m into anchor;
            else
                Create new node/edge for m in 𝒢;
            end
        end
    end
    // Logic Audit and Augmentation (7-dim rules)
    𝒢 ← Agent_ont(𝒢) ;                    // Resolve contradictions and infer new triples
end
return 𝒢;
```